

Loan Approval Prediction System - Round 2

(Algorithm & Dataset Format)

A1. Step-by-Step Procedure

1. Data Collection: Collect the Loan Prediction dataset from Kaggle.
2. Data Loading: Import the CSV dataset using Python and Pandas.
3. Data Cleaning: Handle missing values, remove duplicates and inconsistent records.
4. Feature Selection: Select relevant attributes such as ApplicantIncome, LoanAmount, Credit_History, Education and Property_Area.
5. Data Encoding: Convert categorical values into numerical values using Label Encoding.
6. Data Preparation: Separate input features (X) and target variable (Loan_Status).
7. Train-Test Split: Divide the dataset into 80% training and 20% testing.
8. Model Selection & Training: Train a Logistic Regression model.
9. Prediction: Predict whether the loan will be Approved or Rejected.
10. Evaluation: Evaluate using Accuracy, Precision, Recall and F1-Score.

A2. Logic / Method Used

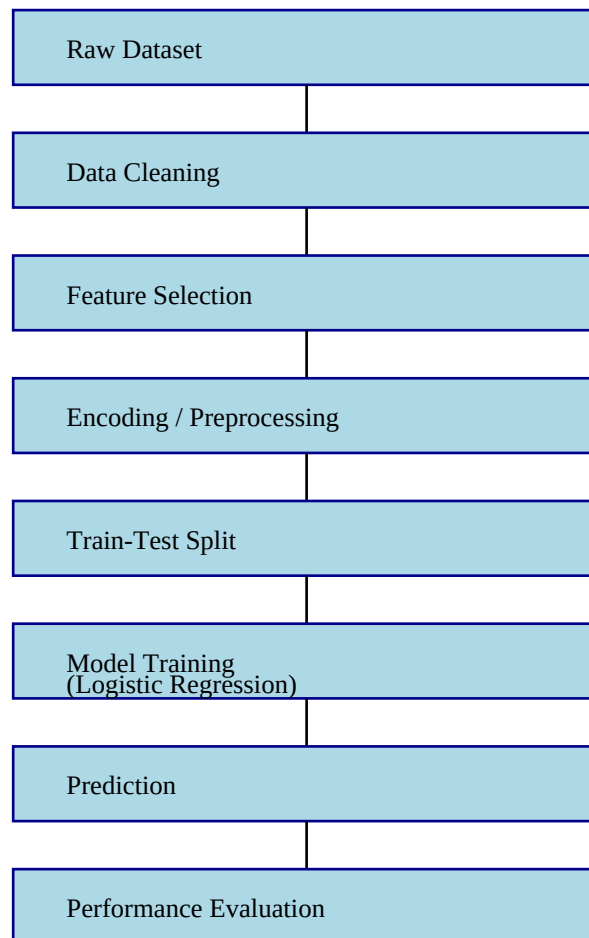
This project is a binary classification problem. After preprocessing and encoding the dataset, Logistic Regression is trained to identify patterns between applicant information and loan approval status. The trained model predicts whether a new applicant is eligible for loan approval.

A3. Model / Technique Applied

Model: Logistic Regression.

Reasons: Simple to implement, suitable for binary classification, fast to train, provides good baseline accuracy, and is widely used for loan approval prediction.

A4. Process Flow of the System



Part B - Dataset Information

Field	Details
Source of Data	Kaggle - Loan Prediction Dataset
File Name/Type	loan_prediction.csv (CSV)
Number of Records	614
Number of Columns	13
Features Used	Gender, Married, Dependents, Education, Self_Employed, ApplicantIncome, CoapplicantIncome, Loan_Amount_Term, Credit_History, Property_Area, Loan_Status
Target Variable	Loan_Status
Train-Test Split	80% Training / 20% Testing
Data Format	Structured CSV Dataset